

Taller 5 - MongoDB con Docker y Docker Compose

Parte 1:

```
juanc@JHANKPC:~/cloud/taller5$ docker compose ps
NAME                IMAGE             COMMAND                                           SERVICE  CREATED
STATUS             PORTS
taller5-mongo-1     mongo:8          "docker-entrypoint.s..."                     mongo    About a minut
e ago              0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
juanc@JHANKPC:~/cloud/taller5$ docker compose exec mongo mongosh -u admin -p
admin --authenticationDatabase admin
Current Mongosh Log ID: 6a9e07a576e202a8ade5b676
Connecting to:      mongodb://<credentials>@127.0.0.1:27017/?directConne
ction=true&serverSelectionTimeoutMS=2000&authSource=admin&appName=mongosh+2.
10.0
Using MongoDB:      8.3.8
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to
MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-09-07T00:37:39.090+00:00: Using the XFS filesystem is strongly recom
mended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-07T00:37:40.232+00:00: For customers running the current memory a
llocator, we suggest changing the contents of the following sysfsFile
2026-09-07T00:37:40.233+00:00: For customers running the current memory a
llocator, we suggest changing the contents of the following sysfsFile
2026-09-07T00:37:40.233+00:00: We suggest setting the contents of sysfsFi
le to 0.
2026-09-07T00:37:40.233+00:00: We suggest setting swappiness to 0 or 1, a
s swapping can cause performance problems.
-----
test> █
```

```

test> use tienda
switched to db tienda
tienda> db.createCollection("products")
{ ok: 1 }
tienda> db.products.insertMany([
|   { name: "Laptop", price: 3500000, stock: 5 },
|   { name: "Mouse", price: 80000, stock: 20 },
|   { name: "Keyboard", price: 150000, stock: 12 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a9e07f076e202a8ade5b677'),
    '1': ObjectId('6a9e07f076e202a8ade5b678'),
    '2': ObjectId('6a9e07f076e202a8ade5b679')
  }
}
tienda> db.products.find()
[
  {
    _id: ObjectId('6a9e07f076e202a8ade5b677'),
    name: 'Laptop',
    price: 3500000,
    stock: 5
  },
  {
    _id: ObjectId('6a9e07f076e202a8ade5b678'),
    name: 'Mouse',
    price: 80000,
    stock: 20
  },
  {
    _id: ObjectId('6a9e07f076e202a8ade5b679'),
    name: 'Keyboard',
    price: 150000,
    stock: 12
  }
]

```

Parte 3:

```

juanc@JHANKPC:~/cloud/taller5$ mkdir -p mongo/init
juanc@JHANKPC:~/cloud/taller5$ nano mongo/init/01-init.js
juanc@JHANKPC:~/cloud/taller5$ nano docker-compose.yml
juanc@JHANKPC:~/cloud/taller5$ nano docker-compose.yml
juanc@JHANKPC:~/cloud/taller5$ tree
.
├── docker-compose.yml
├── mongo
│   └── init
│       └── 01-init.js
3 directories, 2 files

```

```

GNU nano 8.7.1                                docker-compose.yml
services:
  mongo:
    image: mongo:8
    environment:
      MONGO_INITDB_ROOT_USERNAME: admin
      MONGO_INITDB_ROOT_PASSWORD: admin
    ports:
      - "27017:27017"
    volumes:
      - mongo_data:/data/db
      - ./mongo/init:/docker-entrypoint-initdb.d:ro

volumes:
  mongo_data:

```

Parte 4:

```

juanc@JHANKPC:~/cloud/taller5$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED
STATUS             PORTS
taller5-mongo-1     mongo:8           "docker-entrypoint.s..." mongo       8 seconds ago
Up 7 seconds       0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
juanc@JHANKPC:~/cloud/taller5$ docker compose exec mongo mongosh -u admin -p
admin --authenticationDatabase admin
Current Mongosh Log ID: 6a9e09bcaad1c3d834101ebd
Connecting to:         mongodb://<credentials>@127.0.0.1:27017/?directConne
ction=true&serverSelectionTimeoutMS=2000&authSource=admin&appName=mongosh+2.
10.0
Using MongoDB:         8.3.8
Using Mongosh:         2.10.0

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The server generated these startup warnings when booting
2026-09-07T00:47:46.837+00:00: Using the XFS filesystem is strongly recom
mended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-07T00:47:47.942+00:00: For customers running the current memory a
llocator, we suggest changing the contents of the following sysfsFile
2026-09-07T00:47:47.942+00:00: For customers running the current memory a
llocator, we suggest changing the contents of the following sysfsFile
2026-09-07T00:47:47.942+00:00: We suggest setting the contents of sysfsFi
le to 0.
2026-09-07T00:47:47.942+00:00: We suggest setting swappiness to 0 or 1, a
s swapping can cause performance problems.
-----
test> █

```

Parte 5:

```

switched to db tienda
tienda> show collections
customers
orders
products
tienda> db.products.find()
[
  {
    _id: ObjectId('6a9e0b3adab33d221aa66974'),
    name: 'Laptop',
    price: 3500000,
    stock: 5
  },
  {
    _id: ObjectId('6a9e0b3adab33d221aa66975'),
    name: 'Mouse',
    price: 80000,
    stock: 20
  },
  {
    _id: ObjectId('6a9e0b3adab33d221aa66976'),
    name: 'Keyboard',
    price: 150000,
    stock: 12
  }
]
tienda> db.customers.find()
[
  {
    _id: ObjectId('6a9e0b3adab33d221aa66977'),
    id: 1,
    name: 'Ana',
    email: 'ana@example.com'
  },
  {
    _id: ObjectId('6a9e0b3adab33d221aa66978'),
    id: 2,
    name: 'Luis',
    email: 'luis@example.com'
  }
]

```

Parte 6:

1. ¿Qué diferencia existe entre tabla y colección?

Una tabla tiene una estructura definida por columnas, mientras que una colección almacena documentos que pueden tener estructuras diferentes.

2. ¿Qué diferencia existe entre fila y documento?

Una fila representa un registro siguiendo las columnas de una tabla. Un documento almacena información en formato similar a JSON y puede contener estructuras más complejas.

3. ¿Qué ventaja ofrecen los documentos anidados?

Permiten almacenar información relacionada dentro de un mismo documento, reduciendo la necesidad de hacer relaciones o consultas entre varias estructuras.

4. ¿Qué riesgos tiene duplicar datos en documentos?

Puede generar inconsistencias y dificultar las actualizaciones si el mismo dato está almacenado en varios documentos.

5. ¿Qué tipos de aplicaciones podrían beneficiarse de MongoDB?

Aplicaciones con datos flexibles o cambiantes, catálogos, comercio electrónico, aplicaciones web y sistemas que manejan documentos con estructuras variadas.

6. ¿Qué modelo resulta más adecuado para el ejercicio universitario y por qué?

MongoDB resulta útil para este ejercicio porque una orden puede contener sus productos directamente como documentos anidados, mostrando claramente las ventajas del modelo documental.